

Run 12 | NME

India

IV + controls

Target Necessity-motivated entrepreneurship; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 1,849 (6.71%) **Case time** 1m 16s
Train / val. / test 18,740 / 3,305 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9881). Highest F1: XGBoost (0.7809). For XGBoost, adding controls changed ROC-AUC by -0.0002 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9881	0.8266	97.01	76.56	79.67	0.7809	<1s
H2O GBM	0.9879	0.8287	96.77	74.81	78.05	0.7639	3s
CatBoost	0.9879	0.8232	96.48	70.11	82.66	0.7587	14s
Random Forest	0.9871	0.8179	96.41	68.87	84.55	0.7591	1s
Lasso	0.9866	0.8053	96.41	69.39	82.93	0.7556	16s
Elastic Net	0.9866	0.8054	96.46	69.59	83.74	0.7601	17s
Logistic Regression	0.9863	0.8029	96.66	73.90	77.51	0.7566	2s
Ridge	0.9859	0.8073	96.64	73.47	78.05	0.7569	14s
Decision Tree	0.9857	0.7648	96.68	76.27	73.17	0.7469	<1s
Naive Bayes	0.8883	0.4514	93.34	50.24	56.91	0.5337	6s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	649.49	41.55
2	Established business owner	244.83	15.66
3	Occupational status	72.03	4.61
4	Age	62.91	4.03
5	Household size	41.78	2.67
6	Knowing entrepreneurs	33.93	2.17
7	Household income	29.74	1.90
8	Survey year	28.70	1.84

Share is the percentage of total forest impurity importance across all 53 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Freedom to trade internationally	12.95	0.83
2	Sound money	12.33	0.79
3	Legal system and property rights	10.76	0.69

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 12_obj3_india_iv_cv_NME. Scores and importance are taken directly from saved CSV outputs.